Zachary Skalski-Fouts

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EDUCATION

Florida State University | Tallahassee, FL **May 2028**

B.S. in Computer Science, Minors in Mathematics | GPA: 3.97/4.00

- **Graduate Coursework:** Advanced Operating Systems (COP5611), Artificial Intelligence (CAP 5605)
- **Relevant Courses:** Operating Systems, Computer Networks, Computer Organization, Distributed Systems, Data Structures, Algorithms, and Generic Programming I/II, Advanced Java Programming
- **Honors:** President's List (2), Dean's List (2)

TECHNICAL SKILLS

Languages (by proficiency): C++, C, HTML/CSS, JavaScript, Java, Python, MIPS Assembly
Tools and Technologies: Linux, Git/GitHub, Jira, WebSockets, multithreading, synchronization, VirtualBox, JSON

RELEVANT EXPERIENCE

Florida State University Engage 100 Program | Tallahassee, FL **Aug. 2026 - Present**

Computer Science & Engineering Colloquium Instructor

- Selected to independently lead two Computer Science and Engineering colloquium sections, mentoring up to 30 first-year students, and monitoring academic engagement and student progress.
- Collaborated with faculty, guest speakers, and peer leaders to deliver career programming on internships, resumes, interviews, research, and engineering pathways.
- Designed and facilitated interactive lessons on responsible AI use, academic integrity, and technical career preparation, strengthening students' communication, teamwork and problem-solving skills.

Florida State University Department of Computer Science | Tallahassee, FL **October 2025 - July 2026**

Undergraduate Research Intern, Networked AI Systems Lab

- Co-authored two AI-systems papers, including a published multi-agent behavioral-health platform and AeroKV, a resilient distributed LLM-inference architecture for UAV swarms submitted to IEEE MILCOM 2026; contributed to technical writing and revision.
- Co-developed AeroKV's boundary KV-cache snapshot mechanism, enabling UAV swarms to resume distributed inference after aircraft loss while limiting memory, communication, and battery overhead.
- Designed metrics for a behavioral health agent's longitudinal progression and cross-session continuity, measuring repetition, constructive revisiting, actionable progress, and retention of prior goals and barriers

Other Experience: Publix Deli Clerk (June 2024 – September 2024), Ben & Jerry's (December 2021- June 2024), Jeff Ellis Lifeguard (August 2021 – March 2022), High Sierra Pools Lifeguard (May 2021 – August 2021)

PROJECTS

Missile Defense Simulator | (C++, HTML/CSS, JavaScript) | *GitHub* **June 2026 - Present**

- Applied the software development lifecycle through planning, Jira requirements tracking, system design, and implementation for radar, command-and-control, launcher, interceptor, and threat components.
- Engineered a multithreaded backend using Boost.Asio/Beast, WebSockets, thread-safe queues, and JSON messaging to coordinate real-time simulation state between modular components and the operator interface.
- Built and integrated a real-time visualization interface using JavaScript, HTML, and CSS, while managing dependencies and reproducible Windows builds with CMake, vcpkg, and Git.

Linux Elevator Kernel Module | (C) | *GitHub* **Nov. 2025**

- Developed a custom Linux Kernel module using C to implement an elevator scheduling system
- Added new system calls to the kernel, demonstrating proficiency in compiling, configuring, and debugging Linux
- Applied concurrency control using mutexes and kernel threads to manage shared resources

MIPS 5-Stage Pipeline | (C) | *GitHub* **April 2025**

- Developed a cycle-accurate simulator for a five-stage MIPS processor pipeline, modeling instruction fetch, decode, execution, memory access, and write-back.
- Implemented hazard detection, stalling, forwarding, branch handling, and tests for pipeline correctness.